

Understanding Content Creators' Struggles and Expectations of AI in Direct Messaging

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Abstract

Social media content creators engage with their audiences through direct messaging to build self-branding and monetize their content. While private instant messages foster personalized interactions with audiences, creators often face challenges managing a high volume of messages from diverse contexts, leading to relational labor struggles. Through semi-structured interviews with 15 content creators, we explored the challenges they experience in interpreting messages and writing responses. We also explored expectations for AI-mediated communication tools in future designs that facilitate communication between creators and audiences. Our findings reveal that creators often struggle with understanding the message senders' contexts from messages, performing personas, and maintaining personal boundaries in writing their responses. AI support is seen as beneficial, yet there are concerns about its negative effects on maintaining relationships. We propose further research to enhance the design of AI-mediated communication tools.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

content creator, audience, direct message, relational labor

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1 Introduction and Related Work

Social media content creators strategically engage with audiences to build self-branding and monetize their content on online platforms, like replying to comments and sending direct messages [9, 13, 19, 42]. These interactions are expected to capture the attention of the audience and influence the visibility of algorithms or brand campaigns [11, 18]. On the other hand, they sometimes feel

compelled to maintain intimacy as a relational labor, to maintain attention from audiences [8].

One of the activities performed to engage with audiences is 'direct messaging' [35], which refers to communication behaviors via private message chats in platforms. Direct messaging enables creators to interact directly with their audiences, such as brand-company clients and followers. It can create unique communication dynamics different from communication in personal relationships, where creators concurrently exchange messages with multiple people who have largely unknown backgrounds [36]. However, this dynamic often makes it difficult for creators to manage the large volume of independent interactions from various contexts [8, 26]. The HCI community has investigated how platform features, its governance, and audience influence creators' behaviors, emphasizing their needs to sustain growth [18, 32, 33]. However, the specific challenges that creators face when they manage an overwhelming influx of direct messages (DMs) remained underexplored. AI features (e.g., Google Smart Replies [28], Meta AI [34]) can be designed to effectively and efficiently communicate with audiences [24]. Yet, it is still unclear how AI technologies can be conducive in their social context. Understanding creators' current practices can help inform the design of user-centered systems that incorporate their needs and reflect social media platforms' approaches [6, 34] supporting direct messaging.

Through semi-structured interviews with 15 content creators, we answer the following research questions: 'What challenges do creators experience in engaging with audiences through direct messaging?' and 'What expectations and concerns do they have about AI-mediated communication (AI-MC) tools that support creator-audience communication?' The findings indicate that creators face challenges in establishing relationships with key audience groups while maintaining their personal boundaries (Figure 1). They recognized the potential of AI applications, such as suggesting or automating responses, to alleviate the effort required for communication. However, creators remained concerned about the risk of AI-driven miscommunication fostering inauthentic relationships. We discuss potential future work that could expand the design space of AI-MC tools to support communication between creators and audiences.

1.1 Content Creators' Practices to Engage with their Audiences

Engaging with audiences via social media is one of the imperative activities for social media content creators to continue their creative activities on platforms and ground fanbase [13, 19, 39, 46]. This not only fosters long-term fan relationships but also boosts creators' earning potential by influencing their visibility and thus, the

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frequency of algorithmic recommendations and subsequent view counts [9, 18]. This leads them to shape communication strategies focusing on target audiences [17]. For example, they consider the feedback of the audiences in their creatives and use words that their fans can understand [42]. Also, strategic use of direct messages (DMs) and reactions (e.g., likes, comments) across platforms can strengthen creator-audience relationships by demonstrating authenticity and building rapport [5].

However, individual creators often bear the full responsibility for all communication activities. Algorithmic platforms prioritize audience preferences, often pushing creators to engage in various activities to maintain ‘good’ relationships with their audiences [12, 20]. It is estimated that creators often exceed their working hours managing their audiences, while the labor may not be compensated [19]. In a Deloitte survey exploring creators’ needs for Generative AI (GenAI), 23% of respondents expressed a need to automate audience interactions [29]. Researchers have suggested handling massive amounts of audience interactions, such as extracting essential information to get informed of the audience sentiment and how to behave in the space [31, 37]. Jhaver et al. [26] designed a word filter tool with which creators can filter and navigate undesirable comments based on creator-designated rules. However, those approaches may not effectively support dyadic interactions between two individuals. On the other hand, platforms like Instagram have recently tested adopting Large Language Models (LLMs) to quickly respond to fans, analyzing the specific language style and tone of creators’ posts [27, 34]. However, it is less explored how creators perceive these supports and what kind of support they need.

1.2 Direct Messaging on Social Media

‘Direct messaging’ is defined as a communication activity between users utilizing chat message features in online social media platforms [35]. Users privately connect with others through synchronous chats on platforms (e.g., Instagram DM), regardless of their personal familiarity [44]. Such high accessibility can effectively deliver messages that target specific audiences. For example, a user can sell products with DMs to a specific user [16].

On the other hand, users can be susceptible to scams and harmful content from unidentified senders. Users can easily fall victim to scams through personalized and targeted messages [43]. Previous research has mostly focused on how general social media users, especially youth, use direct messages with an emphasis on designing for online safety [7, 25, 38, 44]. Ali et al. [7] highlighted that unsafe and provocative conversations that create distressful emotions are shared through direct messages. Also, Verbeij et al. [44] found that adolescents share happiness more than sadness between peers. Although audiences can easily contact creators via social media DMs, the social contexts in which creators interact with audiences through direct messaging are not well understood.

2 Methodology

We conducted semi-structured interviews with 15 social media content creators over Zoom between October and November 2024. Our research is approved by our institution’s IRB.

2.1 Recruitment and Participants

We used purposive sampling [21] to recruit participants who met the following criteria: 1) self-identify as a content creator with at least 10,000 (10k) followers on multiple social media platforms; 2) regularly receive and send direct messages for professional purposes via social media platforms (e.g., Direct Message from Instagram, TikTok, and X). The criteria was determined based on two pilot interviews with two creators, literature review [18], and the feasibility of reaching potential participants. We found that those with fewer followers generally faced fewer challenges in managing high volumes of direct messages. This follower number threshold helped us focus on creators more likely to encounter significant communication burdens. In the recruitment process, we aimed to balance the topics of their channels and ensure gender diversity. Participants were recruited through creators related Subreddit channels, Prolific, and cold email outreach to eligible creators. Table 1 presents demographic information. 14 participants used Instagram DM and TikTok mainly, and 3 participants used X (formerly Twitter).

2.2 Procedure

The semi-structured interview study consisted of two phases: 1) examining participants’ current direct messaging practices, and 2) exploring their responses to AI-suggested messages in hypothetical scenarios.

For the first phase, drawing upon prior work that focused on handling massive content from audiences [26, 37], we conceptualized direct messaging as two sub-processes: 1) interpreting the sender’s messages and 2) writing responses. We began by asking participants about the common messages they receive and the message senders. Then, focusing on each sub-process, we asked about their decision-making process and any challenges encountered. With participants’ consent, they were encouraged to open their DM inboxes and provide examples from their message history.

In the next phase, we examined participants’ perspectives on technologies that could enhance their direct messaging process. Participants were asked to share their thoughts about scenarios where they respond to audience messages using AI-generated suggestions. We focused on understanding their views on personalized AI writing assistants capable of generating message responses tailored to their channel themes and typical language styles of their channels [22, 34]. Participants were first asked to envision themselves using an AI feature that recommends potential replies to respond to the given messages. Then, we presented exemplary cases of AI-suggested messages tailored for both informal and formal communications. This approach assumes varying needs for AI-suggested messages in these contexts, as indicated by prior research [22]. Three types of messages were generated for each case, considering 1) generic communication, 2) creator information, and 3) the message sender information (Details are described in Appendix A). After reviewing the options, participants were asked to choose one which they would hypothetically use or not, and to share their thoughts and additional suggestions for best practices in direct messaging.

PID	Channel Theme	Gender	# of Followers	Social Media Platforms Where They Use DM	Message Receiving Frequency	Message Replying Frequency
P1	Fashion	M	28k	Instagram, TikTok	4-6 times a week	4-6 times a week
P2	Cryptocurrency	M	14k	Instagram, TikTok, X	4-6 times a week	4-6 times a week
P3	Comedy	M	28.8k	Instagram, TikTok	more than 7 times a week	4-6 times a week
P4	Culture	F	248k	Instagram, TikTok	more than 7 times a week	Once a week
P5	Fashion	F	233.7k	TikTok	more than 7 times a week	2-3 times a week
P6	Shoe-crafting	M	68.8k	Instagram	more than 7 times a week	more than 7 times a week
P7	Music	F	30.5k	Instagram	2-3 times a week	2-3 times a week
P8	Horror	M	302k	Instagram	more than 7 times a week	more than 7 times a week
P9	Lifestyle	M	13k	Instagram	more than 7 times a week	more than 7 times a week
P10	Fashion	F	79.4k	Instagram	4-6 times a week	Once a week
P11	Political news	M	503k	X	4-6 times a week	4-6 times a week
P12	Healthcare news	M	60k	Instagram, X	more than 7 times a week	more than 7 times a week
P13	Cuisine	F	32.2k	Instagram	more than 7 times a week	2-3 times a week
P14	Lifestyle	F	33.1k	Instagram	more than 7 times a week	more than 7 times a week
P15	Dance	M	11.5k	Instagram	more than 7 times a week	2-3 times a week

Table 1: Demographic Information of Participants

2.3 Data Analysis

All interview data was audio recorded and transcribed using Otter.AI [1]. After anonymizing the transcripts, we conducted a thematic analysis [14] using MAXQDA 24.5 [2]. First, two authors independently reviewed the transcripts to gain a comprehensive understanding of the content. Then, each author independently used an open-coding method [45] to descriptively understand how codes were distributed. We then compared our coding scheme, noting similarities and differences between them, followed by memo-writing. After several rounds of discussion, we iteratively refined and merged the codes until we reached an agreement. Finally, guided by our research questions, we identified relationships between codes and synthesized them into higher-level concepts.

3 Findings

3.1 RQ1. Challenges of Engaging with Audiences in Direct Messaging

We identified four themes that illustrate the challenges participants faced in interpreting messages and writing responses.

3.1.1 Interpreting Messages: Assess the Significance of Messages based on the Content of the Messages and Message Senders. All participants stated that they manually skim the content and assess the sender. They prefer to read important messages, like fan compliments or business offers, but current DM inboxes mainly filter out harassment and harmful content, sorting messages from mutual followers into separate inboxes. Therefore, they needed to manually filter out messages containing irrelevant or inappropriate content that was deemed unworthy of a response or made participants feel uncomfortable.

“I’d say the challenge is sometimes it’s (direct messaging) tiresome maybe you can get 100 messages and then they expect you to respond to them. It’s also very streaming. Because someone sent me a very good message, acknowledging me, [...] or maybe giving me a good

advice. But I’m not able to respond to those messages because for maybe I never even saw it.” (P3, Comedy)

DMs deemed significant should contain content relevant to a creator’s posts or channel themes (P1, P4, P5, P7, P11, P12). For example, asking a creator who regularly shares content about Chinese culture for tips on traveling in China would be considered relevant, according to P4. The participants emphasized the importance of DMs that either require immediate action (e.g., finding blood donors) or express deep fandom, characterized by politeness and appreciation. Also, the DM content should not be intended to invade their privacy, by asking about topics outside of the creators’ channels. For example, P1 noted that it is uncomfortable to read messages that senders have sent to creators, which typically contain content suitable only for intimate personal relationships.

Audience groups are also an important factor that participants considered when evaluating the significance of DMs (P3, P6, P7, P9, P12, P15). When the sender is a mutual follower or a potential brand client offering financial benefits, the participants stated that they tried to immediately respond to the DMs to shorten response latency. Through these behaviors, they wanted to construct good relationships with those prioritized groups. P6 mentioned that if he does not respond to the DMs, the senders would view him as unresponsive, potentially damaging his reputation. Messages sent from unknown people and even fans are likely to be handled at the last moment to manage their time and efforts.

3.1.2 Interpreting Messages: Contextualize Message Content to Confirm Authenticity. Message senders may contact creators without prior chat history or use anonymous accounts, making their backgrounds unknown to the creators. Because of this, eight participants (P1, P3 - 5, P7, P9, P12, P13) sometimes view messages from unknown senders with suspicion, worrying that they might be deceptive or pose risks to their privacy and interests. Even when senders offer some business deals or call-to-actions, seven participants reported that they needed more information to understand the authenticity or emotions of the senders. To address this, they

looked for cues to contextualize messages they received, such as confirming verified labels endowed by platforms or if the messages contained content relevant to their posts and channel themes.

“I do check their accounts regularly, because sometimes there are real people who are there. Most of the time I can see accounts trying to do scam activities and dirty messages. With the time I start, I start realizing which kind of messages are real and which kind of messages are not.” (P9, Lifestyle)

Three participants (P1, P4, P7) stated that they utilized email as a formal communication channel in addition to direct message, believing this method to be more credible and authentic. P7 stated that she would reply to a DM when the message sender followed up with an email containing the same content, interpreting this as a genuine effort to reach her through all available channels.

Although few in number, P3 and P12 mentioned that the DMs they received were sometimes difficult to understand because they contained content that required expert knowledge without adequate explanations to aid their understanding. Similarly, P15 was sometimes unsure if the senders were being positive on their channels or attempting to undermine them, as the messages often had a straightforward and impolite tone. These made it difficult to compromise their context to properly answer their messages.

3.1.3 Writing Responses: Adjust the Level of Information According to Audience Groups. Participants (P1, P6, P9, P13, P14) believed that the information shared should vary based on their familiarity and priority on audience groups. They explicitly mentioned ‘personal boundary’, illustrating the unseen boundaries and limits they established with some audience groups, which differed from the casual conversations they had with people in personal relationships (e.g., family). However, P14 mentioned that it was challenging to restrict the message receiver from asking for personal information while maintaining a friendly vibe in their messages.

“I really want to be relatable to my general audience and everyone following me, but the internet is a really weird place, and people start feeling as if they know you that much. [...] Because of that, when I’m interacting with them, I tend to not let them cross a certain line. I want them to know that I’m relatable, but then we are not like.” (P14, Lifestyle)

Participants perceived that high information complexity in their responses might prompt the message receiver to engage in unexpected or unwanted interactions, or even harassment. They believed that writing lengthy, empathetic messages could lead to exhaustion, as it might prompt more interactions with unfamiliar senders. Meanwhile, responding adequately without giving specific information is challenging. Thus, for audience groups with whom they did not wish to establish strong connections, participants often kept their responses brief or reacted with emojis to disengage from messages they considered unnecessary. They believed this approach also helped reduce the number of interactions within a single conversation. For example, P1, P9, and P6 stated that they wrote responses as concise and short, or used emojis as possible, to avoid further receiving questions.

3.1.4 Writing Responses: Perform Personas for Different Audience Groups. Nine participants (P1 - 3, P5, P9 - 12, P14) mentioned maintaining consistent specific characteristics in their written responses (e.g., language tones, styles), which echoes Simpson and Semaan’s findings [42]. They tried to project ‘professional’ and ‘relatable’ personalities that aligned with their channel themes, hoping to be perceived positively by their audience. However, these characteristics did not remain consistent across all message senders in the same way. They sometimes adjusted the tone and details of their responses based on the audience group they perceived the message senders to belong to. For instance, they would categorize message senders into specific groups, such as clients, fans, or unknown individuals. Rather than strictly adhering to their established persona characteristics, they preferred to prioritize their approach based on the audience group, tailoring their responses to align with the traits they deemed most suitable for their prioritized groups.

“I have a different persona from who I actually am. With my followers, I stick with my Tiktok persona. With clients, I’m very professional and straight to the point. Then with my friends and family I’m just casual, whatever. There’s like, three different ways of how I talk to each person” (P5, Fashion)

In this process, seven participants sometimes experienced difficulties in choosing appropriate words and tones that conveyed their intent without offending or upsetting specific audience groups, especially when declining offers (P1, P2, P11, P12), negotiating deals (P3, P5), and consulting the message senders (P9). P1 and P9 expressed a desire to use language that makes them appear open and approachable. However, they found it challenging to decline certain requests, such as sharing personal opinions, while appearing authentic.

“Everyone has their own opinions, and everyone has their own views. So trying to understand each and every person is kind of different. If you ask me the same questions and another one ask me the same question, then I think everyone would expect a different reply, not the same reply.” (P11, Political news)

P10 and P15 stated that they tried to appear straightforward in their responses, because these personalities presented might protect them from being harassed by unknown people. P10 stated that she deliberately used a language tone that appeared ‘sarcastic’ and ‘bitchy’ when responding to DMs. She stated that this method of communication would protect her from sexual harassment while aligning with her content.

3.2 RQ2. Expectations and Concerns of AI-mediated Communication tools

Only five participants had experiences of using AI-powered tools for direct messaging, such as ChatGPT and Grammarly, but all participants desired support from tools that assist communication with audiences with features. While it is not surprising that they wanted automated responses and guidance on what to write, they uniformly shared concerns that the features might not consider their specific contexts, potentially affecting their relationship maintenance.

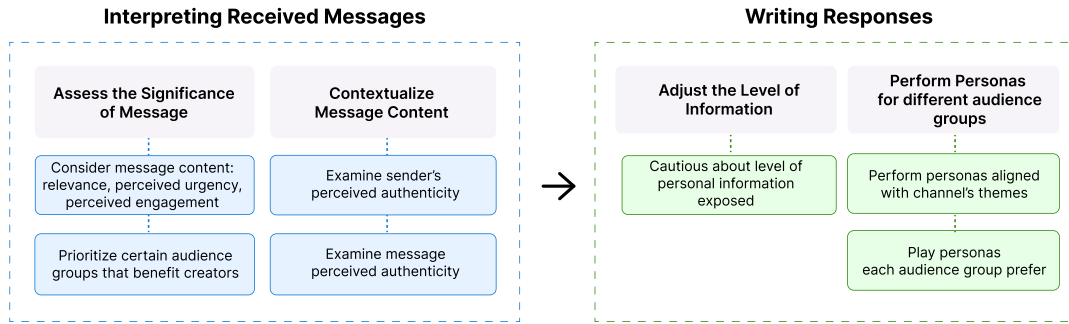


Figure 1: Four Main Challenges and Sub-Challenges Creators Face in Engaging with Direct Messages

Undesirable Content	Desirable Content	
	Automated responses	Guidance in tailoring responses
- Sexually harassing messages (P4, P14) - Hateful content (e.g., negative feedback - P11) - Personal questions (e.g., asking for phone numbers - P1, P9)	- Generic questions (e.g., asking about outfit brands shown in a post - P5) - Recurring messages (e.g., placing an order for shoes - P6) - Brief positive feedback (e.g., 'I like your content' - P1)	- Negotiation (e.g., discussing promotion costs) - Situations that require creators to perform negative politeness (e.g., declining offers - P1) - Messages requiring professional-level domain knowledge (e.g., answering medical-related questions - P12) - Situations where creators need to adopt a more formal tone than usual (e.g., communicating with clients - P9)

Table 2: Types of Message Content Preferred for AI Support and the Desired Forms of Assistance

3.2.1 Automating Responses to Reduce Communication Complexity. Response automation is a widespread approach to managing emails [37] and handling customer support [41], and even commercial DM tools adopt this approach [3]. Similarly, nine participants (P1 - 3, P4, P7 - 10, P12) expressed a desire to automate the entire process of managing direct messages they preferred not to read or respond to manually. These include inappropriate, hateful, or a high volume of repetitive messages from individuals they were not interested in building relationships with. For instance, P9 mentioned that he saw no value in engaging with messages like 'I love you.' Meanwhile, some mentioned that they find it annoying to be ignored after taking the time to respond to messages they find less engaging (P7). Despite their reluctance to even read these messages, participants felt compelled to respond to avoid possible negative consequences.

However, five participants (P1, P3, P7, P8, P10) pointed out that automating direct messaging without specific criteria and errors in simulating their language tones could complicate interactions. P1 and P10 noted that not every message requires a response, as doing so might encourage message senders who intend to bother them. P8 anticipated that the message senders might ridicule or mock him if they discover his responses were automated. P7 also stated that she would like to set static automated responses like

"Thanks for messaging me, I'll respond as soon as I can.", leaving the task of crafting a personalized response to her.

"I think for a vast majority of them (DMs), it's still better to not respond. Because if you don't respond, people can't get offended when you reject them. But if you do respond even if it's the best crafted rejection response in existence, they might still get a negative reaction from it." (P10, Fashion)

Five participants (P1, P3, P4, P6, P15) perceived that automating responses for prioritized audience groups could help reduce delays in communication with these groups. They noted that they frequently missed the opportunities to respond appropriately in real-time, as audiences often discuss repetitive but important topics. Responding in real-time can lead to losing control over their personal lives. For example, P6 perceived that AI agents respond to customers on behalf of him could help place orders quickly. Likewise, during periods of unavailability, automatically generated responses could guide audiences toward further actions to maintain engagement and better meet audience expectations.

3.2.2 Guiding Creators to Effectively Tailor Messages for Relationship Establishment. Six participants (P4, P5, P8, P10, P11, P12) indicated that AI could assist in situations where they struggle to write

responses due to a lack of experiences as professionals. For example, rejecting audiences' offers and persuasion are tricky for some inexperienced creators. Participants wanted to appear professional to the message senders, so they believed that AI assistance could assist them write more professionally in scenarios they typically find challenging. However, they questioned whether AI could fully understand such complex and high-stakes conversations. For example, P12 emphasized the need for accurate information in his channel and expressed concerns that AI might not ensure this accuracy.

On the other hand, P2, P3, and P12 are interested in understanding the level of engagement from the message sender and the contexts in which DMs are written. This context would allow them to personally address sincere and positive messages from the audience. For example, P3 expressed a desire to use AI to analyze her audience's characteristics and reasons for visiting her channels, and then generate suggestions for responses. Participants felt responsible for conveying authenticity in their responses, especially with message senders they regarded as sincere and genuine fans.

However, in these cases of authentic fan engagement, using AI-suggested messages is perceived as less responsible and potentially harmful to the relationship. Participants described AI-generated responses as fostering 'fake and inauthentic relationships', and expressed concerns that errors or missteps in these responses could damage trust and stain their connection with the audiences.

"I'll be willing to use it. But when it comes to sensitive information, like acknowledging my post and also giving me positive energy, I don't want to use the AI to reply to them." (P12, Healthcare news)

Also, P7 and P10 perceived AI-suggested messages that mimicked their language tones as 'overly professional' or 'overly polite.' This indicates that the AI's representation of their styles might not align with their perceived relationships with the message senders.

4 Discussion and Future Work

Previous research [13, 19] has shown that creators build intimacy with their audience by treating them as friends, which adds economic value and aids in self-branding. However, this strategy also involves trade-offs, compelling them to engage in these interactions as a form of relational labor [19, 23]. In line with the prior work, our findings highlight creators' challenges in practicing intimacy through private messages and establishing authentic interactions in numerous message exchanges. These challenges would differ from those in instant messaging platforms (e.g., WhatsApp), which are primarily used for social or work-related personal communication [10]. In social media, audiences who are unlimited and unspecified, can possibly access creators without interpersonal relationships [36]. Findings show that maintaining personal boundaries in self-presentation is essential for creators (Sec 3.1.3, 3.1.4). In this sense, direct messaging can create labor activities influenced by algorithmic precarity [15, 20].

AI support is perceived as helpful for creators in handling messages with varying depths and contexts, but there are suitable and unsuitable scenarios (Table 2) for its use. Also, the extent of AI's supportive role should vary to contribute to relationship maintenance. That is, the effectiveness of AI-MC tools may differ in

relationships compared to interpersonal communication. For example, in interpersonal communication, personal connections are bound in friends, family, or colleges [22], but in creator-audience interactions, the topics and audiences' characteristics may vary depending on creators' channel themes. Also, the communication dynamic is typically one-to-many and creators often adopt a specific persona [42]. This would create a broad communication space where AI-MC tools can be particularly useful, such as tailoring communication styles to suit different target audience groups. Future research could investigate specific scenarios where AI-MC tool is essential and what designs are applicable considering creators' social contexts.

We believe that direct messaging features can be further designed for creators to nurture authentic relationships, reducing uncertainty in interacting with their audiences. In future work, based on the challenges identified through this work, we will explore how Generative AI that considers creators' contexts and personalized interaction impacts creators' direct messaging behaviors with different audience groups.

5 Limitations

We acknowledge several limitations. First, we aimed to garner creators' interaction experiences across diverse audiences through a retrospective interview method [40], which may not present their needs completely. Also, the participants were recruited based on the U.S, which may indicate cultural biases in the results. Additionally, over 80% of participants use Instagram DM for direct messaging, which could introduce biases derived from this single platform. Finally, probes we presented for participants' engagement may not reflect the creators' language tones and styles, depending on the creator data.

6 Conclusion

This study highlights the challenges social media creators face in managing audience interactions through direct messaging, including high message volumes, maintaining authenticity, and navigating diverse contexts. For the use of AI tools for mitigating these burdens, creators express concerns about preserving personal connection and control. By addressing these needs, future solutions could effectively support creators' relational labor while enhancing meaningful engagement with their audiences.

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A Probes

We introduce how we design probes presented in the studies. We first collected participants' public social media posts and direct message cases. Then, we created AI-generated messages through prompts.

A.1 Data Collection

In designing AI-suggested messages used in our studies (§2.2) as probes, we first collected three months of textual content from a participant's primary social media posts, excluding hashtags. Additionally, we gathered direct message examples from social media content creators by utilizing social media posts where creators openly introduced direct messages (10 YouTube videos that contain title "DM" or "Direct Message"). As a result, we collected 203 valid data points. Two researchers sorted the data points as content given

in informal and formal communication situations, based the criteria [22, 30] These messages were used for generating responses in both informal and formal communication with prompt 2(A.2.2).

A.2 Prompt Design

Utilizing the data crawled from participant’s primary social media posts, we aimed to capture participants’ writing styles, including language tones (e.g., encouraging, optimistic), commonly used emojis from social media, and topics they usually write in their social media channel posts (e.g., upcoming events, gratitude towards supporters). Utilizing the data, we extracted the language style information via Prompt 1 in §A.2.1.

We generate three types of AI-generated messages referring to different information: 1) generic communication, 2) creator information, 3) audience information. Generic communication does not consider specific information, instead generating appropriate responses to messages. Creator information-based message considers the extracted information created from Prompt 1. An audience information-based message generates a response based on audience name, following periods and topic of social media content they like. We randomly generate audience information. We generated three AI-suggested messages using a prompt from both informal and formal communication cases after sampling a direct message example (§A.2.2). We employed the GPT-4 API from OpenAI [4].

A.2.1 Prompt 1: Style information.

You are a helpful extractor. We will provide you with the text from a social media creator’s posts. Please extract the language styles present in the text, the frequently used emojis, and the general writing topics covered in the posts. The styles and topics may have multiple attributes.

INPUT: {text}

Output should be a JSON object like this:

Style:

```
- text_styles: [xx, xx, ...],
- emojis: [xx, ...],
- writing_topics: [xx, ...],
- language_tones: [xx, xx, ...]
```

A.2.2 Prompt 2: AI-suggested Messages Generation.

I am a content creator. You will be given below contents: 1. Whether consider My language style ("language-styles":) from my social media posts or not: yes or no
2. The emojis I usually use in social media posts:,
3. Whether consider My writing-topics from my social media posts (topic1, topic2, topic3) or not: yes or no
4. The user name of the audience I want to reply to:
5. How long this audience has been following me, example: 10 years
6. How many times this audience has liked my contents, example: liked 20 posts

7. How many negative comments and positive comments they’ve left on my posts, examples: 20 praises and 0 negative comments

8. What type of content does this audience interact with the most, for example: podcast

9. Message from this audience:

10. Category of the message topic, for example:

Personal questions, Questions/comments/suggestions on contents, Business invitation, Asking for advice for personal issues, Asking for advice for content creation, Random greeting / funny memes, Compliment

11. The language style I want you to reply to message with, for example: informal, neutral, formal

12. How heavily I want you to use emojis in replying to this message, for example: none, low, medium, high

13. How long I want your response to be: For example: short (1-2 sentences), medium (3-4 sentences), long (5+ sentences)

14. Whether or not I want you to label the response as AI-generated: yes or no

You will output THREE replies to 9: For the first reply, only consider (4 & 10 (Choose the right language style for each category) & 14) and generate a response message. For the second reply, only consider (1 & 2 & 3 & 4 & 11 & 12 & 13 & 14) and generate a response message. For the third reply, only consider (4 & 5 & 6 & 7 & 8 & 14) generate a response message. For each reply, also give very short and very precise reasons (can even use key words) after each reply to explain what you consider in the response. If the message contains an Inappropriate/Sexual/Harmful/Money-related spam message, give me THREE replies to let them know politely you cannot respond.